Part 1

Investment Products

CHAPTER 7: Fixed-Income Securities: Pricing and Trading



Chapter Highlights

- Most bonds and fixed-income securities trade in the market on the basis of their yield or yield to maturity
- Changes in interest rates play an important role in the pricing of bonds and fixed-income securities
- The fair price to pay for a bond or fixed-income security is a function of current interest rates, the term to maturity, and the coupon or income component on the security


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Present Value of a Bond

What is the present value of an 8% 6 year semi-annual bond with a $10,000 par value and a 6% discount rate?

Type **12** Press **N**

Type **3** press **I/Y**

Type **400** press **PMT**

Type **10,000** press **FV**

Press **COMP** press **PV**

Answer: **10,995.40**

T-Bill Yields

$$\text{Yield} = \frac{100 - \text{Price}}{\text{Price}} \times \frac{365}{\text{Term}} \times 100$$

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a) What is the yield on a 181 day T-bill purchased for $98,000?

b) What is the yield on a 90 day T-bill purchased for 99?

T-Bill Yields

$$\text{Yield} = \frac{100 - \text{Price}}{\text{Price}} \times \frac{365}{\text{Term}} \times 100$$

a) What is the yield on a 181 day T-bill purchased for $98,000?

$$\frac{100{,}000 - 98{,}000}{98{,}000} \times \frac{365}{181}$$

$$\frac{2{,}000}{98{,}000} \times \frac{365}{181} = 4.12\%$$

T-Bill Yields

$$\text{Yield} = \frac{100 - \text{Price}}{\text{Price}} \times \frac{365}{\text{Term}} \times 100$$

b) What is the yield on a 90 day T-bill purchased for 99?

$$\frac{100 - 99}{99} \times \frac{365}{90}$$

$$\frac{1}{99} \times \frac{365}{90} = 4.10\%$$

Current Yield

Current Yield is a short-term measure of return.

It allows investors to compare investments in bonds with other short-term investment opportunities (GICs, savings accounts, T-Bills etc.)

It is the **actual** interest income rate of return, compared with the coupon rate earned on the bond, based on the prevailing bond price and market interest rate.

$$\text{Current Yield} = \frac{\text{Annual Dollar Amount of Interest}}{\text{Current Market Price}}$$

Current Yield

NFR issued a bond with a 10% coupon, a $1,000 par value and 10 years to maturity.

a) If an investor purchases the bond for $1,000, what is its current yield?

b) If an investor purchase the bond for 90 ($900), what is the investor's current yield?

c) If an investor purchased the bond for 110, what is the investor's current yield?

Current Yield

a) **If an investor purchases the bond for $1,000, what is its current yield?**

$100/$1000 = .10 or 10%

Notice that since the bond sells at par, the coupon and the current yield are the same.

Current Yield

b) If an investor purchases the bond for 90 what is the bond's current yield?

$100/$900 = 11.11%

The price of 90 is simply 90% of par. Since par is $1,000, the price of the bond is: 0.90 × $1,000 = $900.

Why is the current yield higher than the coupon in this case?

Current Yield

c) If an investor purchases the bond for 110, what is the bond's current yield?

$100/$1,100 = 9.09%

Same rule applies here, the purchase price is 110% of par.

Why is the current yield less than the coupon in this case?

Yield to Maturity (YTM)

- The most popular measure of yield in the bond market.
- YTM shows the total return you would expect to earn over the life of a bond starting today.
  - assuming the coupons are reinvested at that YTM
- Reflects the investor's return in the form of coupon income, capital gain (or loss) and receiving par at maturity.
- Best way to solve for YTM is using a financial calculator.

Yield to Maturity

DKE Inc. issues a bond with a 10% annual coupon, a $1,000 par value and 10 years to maturity.

a) What is the bond's YTM if it sells for $1,000?

b) If an investor purchase the bond for 90, what is the bond's YTM?

c) If an investor purchased the bond for 110, what is the bond's YTM?

Yield to Maturity

a) What is the bond's YTM?

- Number of compounding periods = 10
- Present value = 1,000
- Coupon payment = $100
- Future value (Principal) = $1,000
- Solve for YTM = 10%

Yield to Maturity

b) YTM when the bond is purchased for 90

- Number of compounding periods = 10
- Present value = 900
- Coupon payment = $100
- Future value (Principal) = $1,000
- Solve for YTM = 11.75%

Yield to Maturity

c) YTM when the bond is purchased for 110

- Number of compounding periods = 10
- Present value = 1,100
- Coupon payment = $100
- Future value (Principal) = $1,000
- Solve for YTM = 8.48%

Bond Pricing Properties

- Bond prices move **inversely** to bond yields.
- The longer time to maturity the greater the price fluctuation.
- Bonds with higher coupons move less (%) than bonds with lower coupons.
- Special features lead to different pricing (e.g. convertible bonds).
- Bond prices are more volatile when interest rates are low.
- Bonds are subject to reinvestment risk.

Rules for YTM, CY and Coupon Rates

Using what you have learned from the calculations above, lets establish some rules:

a) **If the bond sells at par…**

Coupon Rate = YTM = Current Yield

b) **If the bonds sell at a discount…**

YTM and the Current Yield must be **greater** than the coupon rate.

c) **If the bond trades at a premium…**

YTM and the Current Yield must be **less** than the coupon rate.

Implications: Bond Pricing Properties

Also:

Holding maturities constant, you make more money with a 1% price decline (capital gain), than you lose with the same percentage increase

Bonds with Higher Coupons Move Less (%) than Bonds with Lower Coupons

Example: Two 15-year bonds

Bond C (10% coupon) – 17.3% increase

Bond D (15% coupon) – 16% increase

Implications: Bond Pricing Properties

Bond Prices Move Inversely to Bond Yields

Trading opportunities – "if" you can predict interest rate changes

The Longer to Maturity the Greater the Price Fluctuation

Example: Two bonds with 10% coupon and rates fall to 8%

Bond A (15 years) – $1,172.90 – 17.29% increase

Bond B (25 years) – $1,214.80 – 21.48% increase

Reinvestment Risk

- Since interest rates fluctuate, the interest rate prevailing at the time of purchase is unlikely to be the same as the interest rate prevailing at the time the investor reinvests cash flows from each coupon payment.

- The longer the term to maturity, the less likely it is that interest rates will remain constant over the term. The risk that the coupons cannot be reinvested at the same interest rate that prevailed at the time the bond was purchased is called **reinvestment risk**.

Duration

- A measure of the sensitivity of a bond's price to changes in interest rates.
- It is the approximate percentage change in the price or value of a bond for a 1% change in interest rates.
- The higher the duration of a bond, the more it will react to a change in interest rates.
  - i.e. the higher the duration the riskier the bond

Duration

- Duration is used to determine the volatility or riskiness of a bond or a bond fund.
- If the duration of a bond is 10, then its price will change by approximately 10% if interest rates change by 1%.
- Assume you hold a bond with a 5% coupon that is currently priced at 104.
- If interest rates rise by 1% then the price of the bond will fall by approximately 10% to 93.60.

The Sell Side and the Buy Side

- **Sell side** is the investment dealer side.
  Firms divide fixed-income duties into three roles:
  - Investment banker
  - Trader
  - Sales representative

- **Buy side** is the retail and institutional client side.
  Most buy-side institutional firms divide fixed-income duties into two roles:
  - Portfolio manager
  - Trader

Settlement Exercise

Govt. of Canada T-Bills	Same day
Govt. of Canada bonds (3 years or less to maturity)	T + 2
Govt. of Canada bonds with a term to maturity of more than 3 years and all other bonds, debentures, or other certificates of indebtedness	T + 2